

EXP - 10

nano race_counter.c

```
#include <stdio.h>
#include <pthread.h>
#include <sched.h>
```

```
#define NUM_THREADS 8
#define INCREMENTS 10000
```

```
volatile long counter = 0;
```

```
/* Thread function */
void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        /*
         * Read-modify-write operation is NOT protected.
         * Multiple threads can access counter simultaneously.
         */
        long temp = counter;

        /* Give other threads a chance to run */
        sched_yield();

        counter = temp + 1;
    }

    return NULL;
}
```

```
int main()
{
    pthread_t threads[NUM_THREADS];

    printf("=====\n");
    printf(" RACE CONDITION COUNTER PROGRAM\n");
    printf("=====\n");

    printf("Threads   : %d\n", NUM_THREADS);
    printf("Increments : %d per thread\n", INCREMENTS);
    printf("Expected  : %ld\n",
        (long)NUM_THREADS * INCREMENTS);
}
```

```

/* Create threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    if (pthread_create(&threads[i], NULL,
        increment_counter, NULL) != 0)
    {
        perror("pthread_create");
        return 1;
    }
}

/* Wait for all threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Actual   : %ld\n", counter);

if (counter != (long)NUM_THREADS * INCREMENTS)
    printf("Result   : RACE CONDITION DETECTED\n");
else
    printf("Result   : Expected value obtained\n");

return 0;
}

```

```

yaswanth6970@Yash6970:~/week10$ nano race_counter.c
yaswanth6970@Yash6970:~/week10$ gcc -Wall -Wextra race_counter.c -o race_counter -pthread
race_counter.c: In function 'increment_counter':
race_counter.c:11:31: warning: unused parameter 'arg' [-Wunused-parameter]
11 | void *increment_counter(void *arg)
    |                             ^~~~~
yaswanth6970@Yash6970:~/week10$ ./race_counter
=====
RACE CONDITION COUNTER PROGRAM
=====
Threads : 8
Increments : 10000 per thread
Expected : 80000

Actual : 10320
Result : RACE CONDITION DETECTED
yaswanth6970@Yash6970:~/week10$ |

```

nano mutex_counter.c

```

#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 8
#define INCREMENTS 10000

long counter = 0;

/* Mutex */
pthread_mutex_t lock;

/* Thread function */
void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        /* Lock before accessing shared data */
        pthread_mutex_lock(&lock);

        counter++;

        /* Unlock after accessing shared data */
        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("=====\n");
    printf("  MUTEX COUNTER PROGRAM\n");
    printf("=====\n");

    printf("Threads   : %d\n", NUM_THREADS);
    printf("Increments : %d per thread\n", INCREMENTS);
    printf("Expected   : %ld\n",
        (long)NUM_THREADS * INCREMENTS);

    /* Initialize mutex */
    if (pthread_mutex_init(&lock, NULL) != 0)
    {
        perror("pthread_mutex_init");
        return 1;
    }

```

```

/* Create threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    if (pthread_create(&threads[i], NULL,
        increment_counter, NULL) != 0)
    {
        perror("pthread_create");
        pthread_mutex_destroy(&lock);
        return 1;
    }
}

/* Wait for threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Actual   : %ld\n", counter);

if (counter == (long)NUM_THREADS * INCREMENTS)
    printf("Result   : CORRECT - MUTEX SYNCHRONIZATION\n");
else
    printf("Result   : ERROR\n");

/* Destroy mutex */
pthread_mutex_destroy(&lock);

return 0;
}

```

```

yaswanth6970@Yash6970:~/week10$ nano mutex_counter.c
yaswanth6970@Yash6970:~/week10$ gcc -Wall -Wextra mutex_counter.c -o mutex_counter -pthread
mutex_counter.c: In function 'increment_counter':
mutex_counter.c:11:31: warning: unused parameter 'arg' [-Wunused-parameter]
    11 | void *increment_counter(void *arg)
        |                             ~~~~~^~~
yaswanth6970@Yash6970:~/week10$ ./mutex_counter
=====
MUTEX COUNTER PROGRAM
=====
Threads : 8
Increments : 10000 per thread
Expected : 80000

Actual : 80000
Result : CORRECT - MUTEX SYNCHRONIZATION
yaswanth6970@Yash6970:~/week10$ -|

```

```
yaswanth6970@Yash6970:~/week10$ -echo "===== WITHOUT MUTEX ====="
./race_counter
```

```
echo
echo "===== WITH MUTEX ====="
./mutex_counter
Command '-echo' not found, did you mean:
  command 'aecho' from deb atalkd (4.2.3~ds-2.1ubuntu0.2)
  command 'echo' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
```

```
=====
RACE CONDITION COUNTER PROGRAM
=====
```

```
Threads : 8
Increments : 10000 per thread
Expected : 80000

Actual : 10580
Result : RACE CONDITION DETECTED
```

```
===== WITH MUTEX =====
=====
MUTEX COUNTER PROGRAM
=====
```

```
Threads : 8
Increments : 10000 per thread
Expected : 80000

Actual : 80000
Result : CORRECT - MUTEX SYNCHRONIZATION
```

```
yaswanth6970@Yash6970:~/week10$ |
```